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A mixed-methods evaluation of a statewide naloxone distribution project: considerations for future implementation of opioid overdose prevention initiatives

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Abstract

Background Despite over a 25% national decline in opioid overdose deaths in 2024, overdose remains a leading cause of mortality in the U.S., with persistent regional disparities. Illinois, for example, reported nearly 4,000 overdose deaths in 2022. Evidence-based interventions—such as naloxone distribution—have been integral in reducing fatalities. While naloxone programs vary, research supports broader community access as an effective strategy. In 2021, Illinois launched the *Access* program, a statewide initiative providing free nasal naloxone through an online portal to community organizations, hospitals, and clinics. This study evaluates the implementation and impact of the statewide program to inform future overdose prevention efforts.

Methods A mixed-methods approach was used to evaluate implementation (qualitative) and outcomes (quantitative). Qualitative data included semi-structured interviews ($n = 15$) with state policymakers, implementation partners, and participating organizations. Quantitative data focused on naloxone orders and administrations from two program-level tracking systems. Data were analyzed separately and then compared to interpret findings among the researchers. The evaluation covers three years of data, from July 2021 to June 2024.

Results Quantitative analysis revealed a statistically significant relationship between naloxone orders and administrations, suggesting that increased availability of naloxone in the community is associated with more overdose reversals. Demographic trends showed a rise in administrations on Black individuals and a decline among Hispanic populations. Naloxone use increasingly occurred in residential settings, with a decline in 911 calls and hospitalizations post-administration. Qualitative findings emphasized the importance of no-cost access and a simple ordering process, which allowed organizations to redirect resources to other critical programming needs.

Conclusions The *Access* program successfully expanded naloxone availability across Illinois, particularly among underserved populations. Key facilitators included free ordering and streamlined ordering. However, disparities in naloxone administration responses and institutional participation highlight the need for continued efforts to reduce stigma and promote harm reduction education. Future efforts should focus on expanding uptake in low-barrier

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settings and conducting longitudinal evaluations to assess the sustained impact of naloxone distribution on overdose mortality.

Keywords Naloxone distribution, Harm reduction, Public health, Opioid overdose prevention, Access to care, Mixed-methods evaluation

Background

The number of national opioid overdose deaths lowered in 2024, with a 27% decrease from 2023 (110,037 versus 80,391) [5]. While this decline is promising, overdose remains among the top three causes of death in the United States [1]. Additionally, regional disparities persist, with several midwestern states ranking among the top ten in terms of overdose deaths. For example, Illinois reported over 3,500 overdoses in 2023, with 27% resulting in death [6].

The progress in reducing overdose deaths is largely attributed to the adoption of evidence-based strategies, including public messaging campaigns, prescriber education and tracking systems, medication for opioid use disorders, and naloxone, a medication that reverses opioid overdoses (Massachusetts Health Officers Association [MHOA], [26]). Given the proven effectiveness of naloxone [5], distribution programs have expanded nationwide. These programs vary in scope [25], with some targeting broad community access (e.g., bystanders, businesses) and others focusing only on high-need groups (e.g., people who use drugs, emergency responders) [4]. Given this variation and the critical role of naloxone in reducing overdose deaths, evaluating the implementation and impact of different distribution models is essential [4].

A growing body of research has examined the impact of widespread naloxone distribution programs, aiming to understand whether broader community access to bystanders, families, and businesses can save more lives than distribution limited to emergency responders or high-risk groups [4, 11]. Fischer and colleagues' [11] systematic review and meta-analysis revealed that 44 studies on naloxone distribution programs are effective in increasing survival rates after an overdose. Additionally, a study designed to inform agent-based models on community-based naloxone distribution methods reported that distributing naloxone directly to persons using drugs and other persons in the community (bystanders) reduces the risk of opioid deaths [24]. Despite this evidence, implementation varies widely across states, and more research is needed to understand how different models perform in diverse contexts.

In 2021, Illinois launched a community access naloxone distribution program (herein referred to as '*Access*'), offering free nasal naloxone to community organizations, hospitals, and clinics via an online portal [18]. The *Access* program removes cost barriers to naloxone and

mandates the reporting of naloxone distribution and use [19]. Building on 2010 legislation that allowed for the nonmedical administration of naloxone, Illinois created the Drug Overdose Prevention Program (DOPP), which encourages organizations to implement Opioid Education and Naloxone Distribution (OEND) services. DOPP enrollment is required to participate in the *Access* program [21]. While Illinois has multiple opioid-related initiatives, this study focuses on *Access*, examining its implementation, impact, and lessons learned to inform future efforts.

Purpose

The State of Illinois partnered with the University of Illinois Chicago to evaluate the *Access* program. The study aimed to (1) characterize the organizations and individuals who received naloxone, (2) examine the relationship between naloxone distribution and administration, and (3) explore program implementation successes and challenges.

Methods

The paper utilized mixed methodologies to report on implementation and process outcome data (the quantity of naloxone orders and administrations) and semistructured qualitative interview data (from interviews with state policymakers, implementation partners (i.e., agencies and individuals working at the State level, and participating organizations (i.e., DOPPs and/or OENDs)). A convergent parallel design was used, started with ongoing quantitative data collection; qualitative interviews were then collected concurrently (parallel) with the paper's quantitative data analysis. The quantitative and qualitative data were analyzed separately, with the researchers later comparing the results to interpret the findings [8]. The evaluation spanned from the start of the *Access* project (July 2021) to the end of its third year (June 2024).

Sample and eligibility.

To participate in *Access*, organizations must register as a DOPP, which automatically qualifies these organizations for participation in the project. A select number of DOPPs are also official OEND programs, which receive additional funding from Illinois to provide naloxone education, training, and distribution in their region and to help individuals obtain naloxone. While Illinois has multiple opioid-related initiatives, this sample only includes data points from the *Access* program.

Data collection

Quantitative data collection

This study had two sources of quantitative data collected from July 2021 to June 2024, the time period of the Access program analyzed for this study. The first source came from the distributor of naloxone, and the second came from the Illinois Saves Overdose (naloxone administration) website. The distributor data offered descriptive findings on who (organizational description), how much (amount and cost of the number of boxes containing two 4 mg doses of naloxone ordered), and where (zip code) the naloxone was requested; the dataset was shared with the research team upon request. The naloxone administration data included information exported from the Illinois Saves Overdose database, which provided details on the administered naloxone dosage, the location and timing of administration, and individual-level characteristics of those who received naloxone for attempted opioid reversal. Data entry on naloxone administrations represents input from administrators and employees from DOPP and/or OEND sites to record the number of administrations their agency has made or a report from an enrolled client that their naloxone was used. The dataset on naloxone administrations was accessed via an export function from a report tool. The quantitative portion of this study was approved by the Institutional Review Board at the University of Illinois Chicago.

Qualitative data collection

Fifteen 60-min Zoom interviews were conducted between June 24, 2024, and August 9, 2024, including 5 interviews with policymakers and implementation partners, 5 interviews with OEND program representatives, and 6 interviews with DOPP enrollee representatives (note: one participant was both a policymaker/implementation partner and DOPP enrollee representative). The research team produced a list of 26 outreach candidates that included programs with the highest number of orders from the Access site since July 2021 and incorporating recommendations from Access program leadership regarding which agencies were heavily involved in the Access program; the goal was to recruit a diverse sample of organizations across geography, urban/rural setting, size of agency, and DOPP and non-DOPP enrollees. Beginning in June 2024, the research interview team-initiated email outreach to the candidates, and the participants selected a time using a calendar link. If an interview was not scheduled within seven days of the initial email, the research team sent a reminder email to all nonrespondents. State program leadership prompted unresponsive candidates to participate in an interview. The participants did not receive compensation for participating.

The research team developed semi-structured interview guides tailored to the three key informant roles (see supplemental file). The interviews examined what worked well during implementation, challenges in ordering and distributing naloxone in communities, potential outcomes of the program, and the sustainability of naloxone distribution efforts. Two research team members attended each interview, with one conducting the interview and the second taking detailed notes. At the beginning of each interview, the interviewer obtained consent to participate and recorded the call from the participant. The qualitative portion of the study was approved by the Institutional Review Board at NORC at the University of Chicago.

Data analyses

Quantitative data analysis

Univariate statistics were conducted to describe variables related to the ordering and administration of naloxone. These results are presented as counts and percentages. Concerning multivariate analyses, monthly naloxone administrations were modeled using a negative binomial regression to account for overdispersion of the outcome variable; the variance of naloxone administrations ($1,793 \times 10^8$) exceeded the mean (19,753.33). The estimated dispersion parameter was small but significantly different from zero ($\alpha = 0.019$, 95% CI: 0.0068–0.0525), and the model's likelihood-ratio test of $\alpha = 0$ indicated significant overdispersion (LR chi-bar²(01) = 9.43, $p = 0.001$), supporting the use of negative binomial regression [13, 16].

Predictor variables included the number of naloxone orders and the number of naloxone administrations over the analyzed study period (July 2021–June 2024). The analysis accounts for the time series lag between the variables, as naloxone orders in one month may be used during administration in a later month [29], thus, it includes lag variables for both orders and administrations. Additionally, quarterly indicators were included across the study period months to account for potential season variation. Temporal autocorrelation was assessed using the Durbin-Watson test [9]. The Pregibon [31] link test was used to assess model specifications, the initial models showed minor misspecification ($p = 0.996$). No violations of multicollinearity were found (variance inflation factor [VIF] < 5). Zero-inflated models were not necessary.

Model estimates are reported as incidence rate ratios (IRRs) with 95% confidence intervals (CI). Given the larger counts represented in the naloxone orders, IRRs are interpreted on a meaningful scale (e.g., per 100,000 naloxone orders) to allow for ease of interpretation of effect size. Analyses were conducted via Stata 15 SE.

Qualitative data analysis

To analyze the qualitative interview data, we used an applied thematic analysis approach [15]. First, the interviews were transcribed and uploaded to Dedoose qualitative software. The team developed a codebook concurrently with conducting the interviews. Three team members participated in the coding. To ensure reliability, one team member developed an interrater reliability (IRR) test through Dedoose to ensure alignment in coding as a team. The three coders completed this test and discussed the results. The inter-rater reliability (IRR) from this test ranged from 0.78–0.85, with 1.0 being a perfect alignment (Cohen's Kappa was used to evaluate IRR; Dedoose, [21]). After sufficient IRR was achieved, each interview transcript was then coded by one research team member. After the interview transcripts were coded, the codes were organized and grouped under higher-order headings (i.e., themes), formulating potential subcategories (i.e., subthemes). We also identified representative quotations for themes and subthemes.

Results

Naloxone orders

Over three years, naloxone orders increased from 57,588 cartons in July 2022 to 382,104 in June 2024 (see Table 1). Among the 711,120 total 4 mg cartons ordered during the first three years of the project, the majority were ordered by community and/or social service organizations ($N=406,896$, 57%). Public health departments and other government agencies accounted for approximately one quarter of the total ($N=184,560$, 26%), followed by substance use or mental health service agencies ($N=66,252$, 9%). The remaining orders came from correctional facilities ($N=20,244$, 3%), law enforcement or emergency medical services (EMS)/fire departments ($N=14,244$, 2%), school or educational institutions ($N=12,912$, 2%), and hospitals or Federally Qualified Health Centers (FQHCs; $N=3,612$, 1%).

Qualitative interviews reinforced and contextualized these patterns, highlighting how the program expanded

naloxone access across a wide range of organizations—from small, budget-constrained nonprofits to larger entities such as police departments and colleges. The interviewees emphasized that the program also enabled organizations not previously involved in naloxone distribution to obtain and distribute the medication. Many community-based organizations, particularly OEND programs, use the program's website to place orders on behalf of other groups. As a result, entities such as religious institutions, small businesses (e.g., grocery stores, clothing stores, restaurants, bars, gas stations, funeral homes), libraries, public transportation systems, and mutual aid groups were also able to distribute naloxone within their communities.

“The free access to [naloxone] is provided at such a high level, like such a state level to so many different organizations. The ability to access it does trickle down to other smaller groups and smaller organizations.” – Policymaker/implementation partner

The interviews also revealed that fewer than ten percent of naloxone orders were collectively placed by correctional facilities, law enforcement or EMS/fire departments, school or educational institutions, and hospitals or FQHCs. The interviewees identified several barriers to naloxone distribution in these settings. In schools, nurses and administrators are often hesitant to receive, carry, and distribute naloxone due to uncertainty about how it aligns with existing school policies. In correctional settings, such as jails and prisons, staff were frequently reluctant to provide naloxone to current inmates and returning citizens (people leaving correctional facilities), citing stigma, a lack of awareness, and logistical challenges. In hospital settings, particularly emergency departments, interviewees noted that the program's reporting requirements created an additional burden for already overstretched staff, as the process was separate from standard medical charting and workflows.

Table 1 Naloxone cartons ordered by organizational type, overall and by fiscal year (FY), 2022–2024

Organizational Type	Overall		FY 2022		FY 2023		FY 2024	
	N	%	N	%	N	%	N	%
Total	711,120	100%	57,588	100%	271,428	100%	382,104	100%
Community and/or social service organizations	406,896	57%	36,876	64%	144,060	53%	225,960	59%
Public health departments and other government agencies	184,560	26%	10,896	19%	80,736	30%	92,928	24%
Substance use or mental health services	66,252	9%	6,888	12%	17,784	7%	41,580	11%
Correctional facilities	20,244	3%	420	1%	11,592	4%	8,232	2%
Law enforcement or emergency medical services (EMS)/fire	14,244	2%	432	1%	10,272	4%	3,540	1%
Schools or educational institutions	12,912	2%	2,076	4%	6,480	2%	4,356	1%
Hospitals or Federally Qualified Health Centers (FQHCs)	3,612	1%	0	0%	420	< 1%	3,192	1%
Other	2,400	< 1%	0	0%	84	< 1%	2,316	1%

Cartons included 2 doses of 4 mg/1 ml naloxone nasal sprays. Illinois' fiscal year is July 1 through June 30

“Our sheriff is very resistant. We’ve had a little committee working with them in the jail to try and inform them and help them understand why it was important to... give [inmates] [naloxone] on their way out. The pushback was, “Oh, that’s too much work. We don’t have enough staff.”– OEND program representative

Naloxone administration

Reports of naloxone administrations nearly tripled, from 255 overdose reversals in July 2021 to 909 administrations

in June 2024. [Appendix Table 4](#) shows the monthly number of administrations over the analyzed project periods. In FYs 2022 and 2023, the number of administrations was greater from March through June. In contrast, in FY 2024, the number of administrations was greater from August through October.

Overall, administrations most often occurred in a house ($N = 1,075$; 64.1%), followed by a public space outdoors ($N = 273$; 16.3%) (see [Table 2](#)). The average person receiving naloxone during the project period analyzed was reported as a white male between the ages of 25

Table 2 Characteristics of reported naloxone administrations, overall and by fiscal year (FY), 2022–2024

Total	Overall		FY 2022		FY 2023		FY 2024	
	N	%	N	%	N	%	N	%
	1,678	100%	255	100%	514	100%	909	100%
Gender								
Male	996	59.4%	158	62.0%	353	68.7%	485	53.4%
Female	602	35.9%	79	31.0%	139	27.0%	384	42.2%
Transgender	25	1.5%	1	0.4%	8	1.6%	16	1.8%
Unknown	55	3.3%	17	6.7%	14	2.7%	24	2.6%
Age								
Under 18	8	0.1%	0	0.0%	1	0.1%	7	0.8%
18–24	179	10.7%	22	8.6%	46	8.9%	111	12.2%
25–44	913	54.4%	147	57.6%	275	53.5%	491	54.0%
45–64	397	23.7%	51	20.0%	127	24.7%	219	24.1%
65 +	42	2.5%	1	0.4%	23	4.5%	18	2.0%
Unknown	139	8.3%	34	13.3%	42	8.2%	63	7.0%
Race and Ethnicity								
Black or African American	509	30.3%	36	14.1%	155	30.2%	318	35.0%
Hispanic/Latinx	200	11.9%	46	18.0%	57	11.1%	97	10.7%
White	829	49.4%	126	49.4%	261	50.8%	442	48.6%
Other*	35	2.1%	7	2.7%	15	2.9%	13	1.4%
Unknown	105	6.3%	40	15.7%	26	5.1%	39	4.3%
Location of Administration								
House	1075	64.1%	132	51.8%	296	57.6%	647	71.2%
Other Residence	66	3.9%	7	2.7%	31	6.0%	28	3.1%
Public Building	68	4.1%	7	2.7%	38	7.4%	23	2.5%
Business Site	93	5.5%	12	4.7%	38	7.4%	43	4.7%
Public Outdoor Site	273	16.3%	56	22.0%	82	16.0%	135	14.9%
Unknown	38	2.3%	19	7.5%	10	1.9%	9	1.0%
Other	65	3.9%	22	8.6%	19	3.7%	24	2.6%
Hospitalization								
Yes	554	33.0%	99	38.8%	192	37.4%	263	28.9%
No	921	54.9%	91	35.7%	237	46.1%	593	65.2%
Unknown	203	12.1%	65	25.5%	85	16.5%	53	5.8%
911 Call								
Yes	609	36.3%	135	52.9%	221	43.0%	253	27.8%
No	890	53.0%	71	27.8%	210	40.9%	609	67.0%
Unknown	179	10.7%	49	19.2%	83	16.1%	47	5%
Consciousness								
Yes	111	6.6%	20	7.8%	50	9.7%	41	5.2%
No	1409	84.0%	190	74.5%	411	80.0%	808	88.9%
Unknown	158	9.4%	45	17.6%	53	10.3%	60	6.6%

*Other races/ethnicities include American Indian, Asian, Native Hawaiian or Pacific Islander, and Multi Racial and Ethnic

and 44 years. However, naloxone administrations among Black or African American individuals increased over the course of the analyzed project period, from 14.1% ($N=36$) in FY 2022 to 35% ($N=318$) in FY 2024. Additionally, from FY 2023 to FY 2024, the percentage of males receiving an administration decreased (FY 2023: $N=353$, 68.7%; FY 2024: $N=485$, 53.4%), whereas the percentage of administrations to females increased (FY 2023: $N=139$, 27.0%; FY 2024: $N=384$, 42.2%).

Most people were found unconscious at the time of administration ($N=1,409$; 84%). The number of calls to 911 during administration decreased from 53% in FY 2022 ($N=135$) to 28% in FY 2024 ($N=253$). In FY 2024, the majority of administrations ($N=609$; 67%) did not result in a call to 911. Similarly, hospitalizations after administration decreased across the project years. In FY 2022, 39% ($N=99$) of administrations resulted in hospitalizations, compared with 29% ($N=263$) by FY 2024.

Qualitative interviews illustrated the significance behind these naloxone administration statistics by

offering context and nuance through the experiences, facilitators, and any challenges shared in the ordering and/or distribution process. Specifically, many interviewees shared first- and second-hand stories of reversing overdoses for friends and family members, often in home settings, aligning with the data showing that most administrations occurred in private residences. The interviewees consistently described the positive impact of naloxone administration not only for overdose survivors but also for families and communities, who avoided the emotional and financial toll of losing loved ones owing to timely naloxone use.

“Speaking as a family member, I think that the people who are going to benefit the most are the people that love someone who uses drugs, because they still have them. They didn’t have to bury them.” – DOPP enrollee representative

The interviewees also emphasized that the actual number of naloxone administrations was likely greater than reported. They highlighted significant challenges in collecting data during overdose events. During an overdose emergency, the immediate focus was on saving the individual’s life, not on collecting or reporting the incident, often resulting in incomplete or missing data. Bystanders who were administered naloxone were also often unaware of the reporting process. Privacy concerns and emotional distress also contributed to reluctance among both overdose survivors and responders to share details after the event. While the quantitative data revealed increased administration over time, the qualitative findings suggest that these data likely underestimate the true extent of naloxone administration.

“I can probably count on one hand the number of times that a participant that I’ve worked directly with has called me saying that they used [naloxone]... Since a lot of that [overdose reversal data] is either secondhand or delayed, the accuracy of... reversal entries is lacking at times.” – DOPP enrollee representative

Table 3 Negative binomial results for monthly naloxone administrations

Variable ($N=35$)	Incidence Rate Ratio (IRR) [‡]	95% Confidence Intervals (CI) [‡]	p-value
Monthly			
Naloxone Orders	1.00	1, 1.00	0.05
Lagged			
Naloxone Orders	1.00	.10, 1.00	0.38
Naloxone Administrations	1.00	.10, 1.01	0.25
Quarterly (Qtr)			
July to September 2021 Qtr 1	reference		
August to December 2021	2.28	1.08, 4.80	0.03
Qtr 2			
January to March 2022 Qtr 3	5.43	2.68, 11.02	0.00
April to June 2022 Qtr 4	5.36	2.51, 11.48	0.00
July to September 2022 Qtr 1	4.11	1.89, 8.94	0.00
August to December 2022	3.94	1.82, 8.56	0.001
Qtr 2			
January to March 2023 Qtr 3	4.26	1.76, 10.27	0.001
April to June 2023 Qtr 4	4.76	1.62, 14.01	0.005
July to September 2023 Qtr 1	6.03	1.10, 18.22	0.001
August to December 2023	5.41	1.819, 16.12	0.002
Qtr 2			
January to March 2024 Qtr 3	6.03	2.25, 16.14	0.00
April to June 2024 Qtr 4	3.49	2.44, 11.18	0.04
Model Fit			
R ²	0.254		
Log-Likelihood	−122.981		
AIC	303.681		
Dispersion Parameter (α)	0.019		

Total observations in the dataset is 36, but the model dropped one due to inclusion of lagged variables

[‡]Decimals were reduced in the table for ease of reading, however the true decimals are reported in the narrative

The relationship between naloxone order and naloxone administration

Given the dearth of evidence on State-wide implementation programming for naloxone, it was important to understand the relationship between the number of orders and when naloxone administrations occurred. Multivariable regression analysis results are represented in Table 3.

Results suggest that for every 100,000 orders of naloxone, a 1.7% increase in naloxone administrations each month (IRR = 1.000017, CI: 1, 1.00003, $p=0.05$).

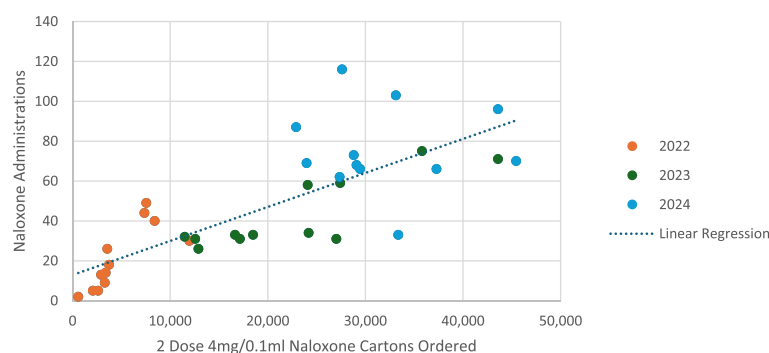


Fig. 1 Number of naloxone cartons ordered and naloxone administered by fiscal month, 2022–2024

Analyses revealed that prior-month orders and administrations (lagged) did not influence current-month administrations. Quarterly indicators showed increases with administrations rising approximately two to six-fold across quarters relative to the start of the project period (IRR: 2.23 in FY22 Quarter 1 to 6.03 in FY 24 Quarter 3, $p < 0.05$); in other words, this suggests that naloxone administration volume is most strongly associated by seasonal ordering activity, that is perhaps influenced by quarterly overdose rates.

The number of naloxone orders explained 25% of the variance in administrations ($R\text{-squared} = 0.254$). Figure 1 shows more variability in the number of administrations at the higher levels of naloxone orders.

Qualitative interviews helped explain why increased distribution may have led to more administrations. The interviewees commonly described how the program's no-cost, easy-to-order naloxone improved access and saturation across communities, making it more likely that naloxone would be available and used during an overdose event. The user-friendly online portal was critical, allowing organizations to quickly place orders and respond to emerging local overdose trends.

“With [naloxone] saturation, one of the other things is the speed to which you can do it. [The program] gives us the speed to be able to quickly respond when we see a spike. I think the speed is one of the biggest things.” – OEND program representative

Additionally, because the program covered the cost of naloxone, organizations could allocate more resources to staffing and outreach-related expenses (e.g., gas, mileage, harm reduction supplies). This enabled organizations to expand their outreach efforts, particularly in high-risk settings and among populations most likely to witness and respond to overdoses. With greater naloxone supply and increased staff capacity, organizations conducted door-to-door distributions and placed naloxone in accessible community locations such as vending machines, repurposed newspaper boxes (e.g., barneys- a newspaper

style naloxone dispenser), front desks, and bathrooms in neighborhoods with high overdose rates. They also distributed more naloxone directly to unhoused individuals living on the street or in homeless encampments. Some organizations have established helplines or websites to help individuals locate naloxone and many paired naloxone distributions with additional harm reduction supplies (e.g., fentanyl test strips) to better engage people who use drugs.

“If we were paying for nasal [naloxone] out of our own pocket, just last month, was the cost of two full-time headcounts for an outreach coordinator. That's incredible. If we had to pay for our [naloxone], we'd still be small. There'd be like four of us... [The program allowed us] not only to expand our staff but to expand our footprint in general. The amount of people that we can reach now because we have more [staff] is tremendous... The state picking up the bill on [naloxone] has allowed us to do some really cool stuff and be able to reach people that historically we've been unable to.” – DOPP enrollee representative

In addition to naloxone administration, analyzing quantitative overdose data was outside the scope of this study. However, several interviewees reported decreases in overdose fatalities and increases in nonfatal overdoses in their program or county-level data—trends attributed to the widespread availability of naloxone. For example, one OEND program representative stated that they “reduced overdose fatalities by 30%” by saturating their county with naloxone. While nonfatal overdoses had increased, “fatal overdoses are dropping, which means we are doing our job.” Although the interviewees acknowledged the lack of concrete, causal data linking the program directly to reduced deaths, many pointed to anecdotal evidence and local overdose trends as strong indicators that the program was contributing to lives being saved.

“Just having this amount of Narcan available, it has a direct correlation with more people surviving. At the end of the day, that’s what this is all about. Because people do recover. People can recover. They just need to be given the chance to do it. Part of that chance is living long enough to be able to do something different. To me, in a nutshell, that is what this program is doing.” – DOPP enrollee representative

A list of parent codes and their corresponding subcodes is provided in Table 5 in [Appendix](#).

Discussion

Qualitative and quantitative data identified key variables contributing to the successful implementation of the statewide *Access* program. Quantitative analyses revealed a positive and statistically significant relationship between naloxone order and administration, suggesting that increased community saturation is correlated with increased overdose reversals. The qualitative data further emphasized the importance of the program’s free access to naloxone and simple ordering process. The interviewees consistently highlighted how these features enabled them to obtain life-saving medication for their communities while also freeing up resources for other critical needs, such as staffing and harm reduction supplies.

The findings also indicated shifts in the demographics of naloxone recipients over the three program years. While white individuals consistently made up the majority of reported recipients, administrations among Black individuals increased over time, whereas the proportion of Hispanic individuals and those classified as ‘unknown’ decreased. These trends likely reflect improved and more consistent data tracking by *Access* program participants. Importantly, these demographic data also underscore ongoing efforts to address racial health disparities in Illinois, where the overdose fatality rate among black residents remains the highest across all racial groups (Illinois Department of Public Health [IDPH], [20]; see Table 3).

Additionally, the data revealed an increase in naloxone administrations occurring in residential settings over the program’s duration. Recent qualitative work from Kahn and colleagues [22] emphasized the importance of “take-home naloxone” initiatives. The participants noted that naloxone availability in the home was often critical for enabling a bystander or loved one to respond effectively during an overdose. However, having naloxone does not benefit individuals who use drugs in isolation, which highlights the ongoing need for harm reduction education to address the risks of using opioids alone [22].

A decline in hospitalizations and 911 calls following naloxone administration was also observed over the program years. The reasons for this decrease may be explained by Kahn et al.’s [22] findings, where 88% of

participants who were treated in a hospital after naloxone administration reported experiencing stigma and shame from healthcare staff and/or emergency responders. In another study, law enforcement officers reported that overdose calls typically result in transportation to either a hospital or jail [34]. As a result, individuals may choose not to seek emergency assistance during an overdose to avoid stigma or potential legal consequences. These patterns may contribute to the observed decrease in emergency responses and underscore the need for more compassionate, coordinated responses that align with Good Samaritan laws.

Relatedly, we found that fewer than ten percent of naloxone orders were placed by correctional facilities, law enforcement or EMS/fire departments, schools, hospitals or FQHCs. Our qualitative data suggest that institutional stigma remains a significant barrier to naloxone distribution in criminal justice, educational, and hospital settings. Moving forward, identifying and leveraging trusted messengers may be an effective strategy to increase naloxone uptake in environments that may be less familiar with or comfortable with the medication [33]. For example, engaging respected team members such as supportive school nurses, sheriffs, or emergency room physicians to advocate for naloxone use with administrators could help reduce stigma and encourage broader adoption across these settings [12].

Limitations

There are limitations to consider for these data and reporting. The information is solely representative of the data associated with the *Access* project; other naloxone access and administrations from other state initiatives are not represented in this study. Given that administration data is self-reported, it is possible that there are mis- or underreports of the number of naloxone administrations. The data and figures reported are reflective of orders made by organizations throughout the State of Illinois and do not necessarily reflect where the orders are then distributed or to whom. Throughout Illinois, there are other avenues to receive naloxone that are not captured here. For example, naloxone can be purchased over the counter. The growing number of options to receive low/no-cost naloxone, coupled with the nationwide problem of mis/underreporting of the administration/overdose reversal counts, creates a challenge to specify and pinpoint the direct impact of the *Access* program on state-wide overdose rates and naloxone distribution. Additionally, it is important to consider the generalizability of the results. The *Access* program is unique to State-level variables and representative of one Midwest State (Illinois). Furthermore, the data are representative of 36 months; thus, more data will be needed to understand the impact of initiatives like this. The smaller sample

size produces smaller effect sizes and should be considered in the interpretation and application of the results. The qualitative component of this study was limited by a small sample size of 15 interviews, which included only a few participants from each stakeholder group and did not include individuals who used drugs or those who had received naloxone through the *Access* project. Despite various state initiatives to provide low- or no-cost naloxone, the anonymity of dispersal and administration often hinders accurate tracking of administrations or how the naloxone orders are distributed by zip code. These are essential factors to understanding saturation and an important future research question that could strengthen opioid overdose prevention literature.

Implications for practice and policy

The findings underscore the effectiveness of harm reduction policies. Specifically, this study found an increase in administrations occurring in homes and residences. Sustaining efforts of existing harm reduction programming could look like community-based agencies distributing naloxone to staff and encouraging them to keep the medication on their person during outreach and field work. Furthermore, continued education on the dangers of using drugs alone will be vital, especially with the rise of fentanyl, which is now found in most drugs being sold [22]. Options to expand existing harm reduction programming include safe consumption sites (SCSs), which provide safe places where individuals feel safe and have access to testing strips, naloxone, and other psychosocial support [37]. A recent systematic review and thematic analysis of SCS research revealed that SCSs foster a sense of inclusion versus stigma while also increasing the public's sense of safety, as it reduces the visibility of drug use in public [37]. Additionally, Yoon et al. [37] reported that persons using drugs will utilize SCSs due to the inclusion of peer workers, which creates a sense of community. With these SCS findings in mind, local legislation and community-based services could promote or support safe consumption sites, decreasing the likelihood of their use alone. Additionally, increased education and dissemination of information about SafeSpot (1-800-972-0590; 2023) could benefit not only those using in isolation, but researchers, clinicians, policy makers, and the general public. SafeSpot is an anonymous hotline that individuals can call before they use and will send for emergency response if individuals stop responding and mitigates the chance of overdose when using alone [32].

These findings also have future policy implications. They suggest that preventive policies and programs such as *Access* should be sustained and additional resources considered. For example, legislation that requires insurance companies to reimburse hospitals for take-home naloxone prescriptions could be implemented. Hospitals

are key providers of emergency medical services, with up to one-third of persons dying of overdoses having accessed emergency services within the year before their death [2]. Additional changes to policies on insurance and access laws could be modeled from some that already exist. For example, Naloxone Access Laws (NALs) were created to decrease barriers to the availability of and access to and availability of naloxone, thereby allowing over-the-counter purchases in pharmacies and distribution through community-based efforts [4]. Although NALs are vital to naloxone availability, there is variability in NALs across the U.S. [4]. Medicaid is a major insurer for substance use services, often being a key reimbursement entity for naloxone distribution and prescription. One study reviewed NALs and their relationship with Medicaid reimbursement and reported consistent associations with increases in naloxone dispensing [14]. These findings highlight an important opportunity for streamlined policy to ensure consistent reimbursement of life-saving medications that can be prescribed over the counter and at hospital discharge, not just during the provision of emergency services (see *The Naloxone Project* in Colorado; naloxoneproject.com). Moreover, future studies could consider a cost analysis study to understand the cost-effectiveness of naloxone distribution and the number of potential lives saved; the New York State Department of Health found a return of investment of over \$3,000 for every \$1 spent on naloxone administration and over 6,500 estimated lives saved [17].

Our results also indicated a decrease in hospitalizations and 911 calls. This aligns with extant research highlighting that even though Good Samaritan laws are designed to protect against bystander intervention, fear of prosecution, arrest, and hesitancy in seeking harm reduction care still exists [10, 22]. To reduce the stigma associated with drug use, continued harm reduction literacy is important for policymakers, law enforcement officials, and providers. Future policy should focus on incorporating health literacy education on the importance of harm reduction integration into micro, mezzo-, and macro-level systems of care, which are inclusive of enhancing knowledge among providers, systems, clients, and the general public [3]. Importantly, best practices for OEND programming include focusing education efforts via outreach in the community, in line with culturally appropriate services; other best practice categories include staff training and support; naloxone saturation and supply; and harm reduction [36]. Decreases in 911 calls and hospitalizations also aligns with evidence that help-seeking behavior is lower among populations using drugs; thus, we must also support creative, albeit controversial, methods to increase the utilization of helping services along the continuum of care. These methods might include implementing "street medicine" and mobile treatment

units that deliver medical care out of mobile vans as they meet people where they are by entering neighborhoods with drug use and overdose hotspots [23]. A recent study implementing a mobile treatment unit alongside their harm reduction outreach efforts offered low-barrier access to needed medical services, including medication for opioid use disorders, and increased follow-up visits from existing clients [28].

While the results cannot directly answer where naloxone saturation occurred in Illinois communities, future research on this topic would substantially benefit opioid overdose prevention literature. As stated in our limitations section, the results of this study are valuable and must be considered in the context of the size of the data; thus, additional research should consider the value of a larger sample size to accurately understand the long-term impact and associations between orders and administrations in state-wide distribution initiatives. Future studies could include a geo-spatial analysis of where (e.g., zip codes) naloxone orders are distributed to versus administered to better understand impacts on the opioid overdose rates. More novel approaches might include methods to better understand the dispersal and saturation of naloxone into different communities by investing in wastewater analysis. Wastewater analysis is currently being used in emerging research in high-overdose areas across diverse counties in the country. The results from longitudinal wastewater data across 40 weeks substantiate the impact this novel methodology has in anticipating new adulterants, such as xylazine, in drug supplies that allow providers and emergency departments to respond better to public health responses [7]. The wastewater data also inform analytics and algorithms to successfully predict drug overdoses up to 11 days in advance [27]. While wastewater analysis has been utilized in major metropolitan areas in the past to understand drug use across the calendar year [7], recent evidence shows that wastewater methods can offer evidence to shape more humane and just drug-related policies. For example, wastewater analysis allowed two counties in Kentucky that decriminalized the use of fentanyl drug testing strips to observe a decrease in fentanyl in their water supply [27]. Although this method is difficult to study, and often anonymous, owing to the nature of naloxone distribution and administration effort, waste water analysis allows us to understand and track where naloxone goes after leaving the hands of outreach workers and providers, as well as the location of where naloxone is administered, which is currently a gap in knowledge [35].

In summary, the study adds to a growing base of evidence indicating that distributing naloxone is cost-effective and saves lives [11, 17]. Further, limiting or removing access to low-barrier naloxone has the potential to reverse the public health improvements and

decreases in national overdose rates, with disproportionate impact on racial/ethnic minorities [30]. As political landscapes change, we must ask ourselves what happens when agencies and communities throughout the United States lack access to Naloxone. The *Access* project was an investment from the State of Illinois in response to the growing number of opioid overdoses prior to the start of the statewide implementation of naloxone. The *Access* project reduced one barrier to ensuring that life-saving naloxone could be ordered, received, distributed, and administered at no cost to community organizations, healthcare organizations, and clinics working with historically underserved, marginalized, and less visible populations (including persons using opioids). Since the implementation of the program, orders for naloxone have increased, as have opioid overdose reversals/naloxone administration. While both naloxone orders and administrations continue, our results suggest that the intended goal of naloxone saturation within at-risk communities is improving, with more naloxone available when overdoses occur. However, more evaluation and outcome research is needed to better understand the longitudinal impact of naloxone distribution versus overdose rates.

Appendix

Table 4 Number of Naloxone Cartons Ordered and Naloxone Administered by Month, FY 2022-2024

Fiscal Month and Year	4 mg/0.1 ml Naloxone Cartons Ordered	Naloxone Administered
FY 2022	57,588	255
July	588	2
August	2,628	5
September	2,088	5
October	3,312	9
November	3,408	14
December	2,928	13
January	3,564	26
February	3,744	18
March	7,560	49
April	7,368	44
May	8,424	40
June	11,976	30
FY 2023	271,428	514
July	11,496	32
August	18,504	33
September	17,148	31
October	16,656	33
November	12,900	26
December	12,576	31
January	24,192	34
February	27,036	31
March	24,096	58

Table 4 Number of Naloxone Cartons Ordered and Naloxone Administered by Month, FY 2022–2024

Fiscal Month and Year	4 mg/0.1 ml Naloxone Cartons Ordered	Naloxone Administered
April	27,420	59
May	43,596	71
June	35,808	75
FY 2024	382,104	909
July	27,372	62
August	43,596	96
September	27,612	116
October	33,132	103
November	29,100	68
December	23,988	69
January	22,908	87
February	28,812	73
March	29,460	66
April	45,456	70
May	37,296	66
June	33,372	33

Table 5 Parent codes and respective sub-codes utilized in the qualitative analysis

Organization background	Project Process	Implementation
Population/area served by organization	Distributing Naloxone	Challenges/unintended consequences
Services provided by organization	Education on Naloxone use	Engagement with Access program leadership
	Ordering through Access website	Positive outcomes/effectiveness
	Receiving Naloxone orders	Recommendations for Access specifically
	Reporting/recording Naloxone distribution	State and federal policy context
		Sustainability
		What worked well

Abbreviations

DOPP	Drug Overdose Prevention Program
OEND	Opioid Education and Naloxone Distribution
IRR	Interrater reliability
EMS	Emergency Medical Services
FHQC	Federally Qualified Health Center
IDPH	Illinois Department of Public Health
SCSs	Safe consumption sites
NALS	Naloxone Access Laws

Supplementary Information

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Supplementary Material 1.

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Authors' contributions

SB was the co-investigator of this project and oversaw the evaluation and reporting for the project; additionally, SB led the writing of the narrative for the background, the quantitative methodology and results, and the conclusion section. OG led the writing of the qualitative results and outlines for the background and conclusion sections. SB and OG collaborated on figure and table creation. AC, NM, and AG also participated in content development and edits for the entire document. OG, AC, NM, and AG were the interview research team overseeing the qualitative portion of this study. KM oversaw the implementation of the *Access* project, contributed to discussion topics, and edited/approved the entire document.

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Data availability

The data analyzed during the current study are not publicly available, as this was not in the permissions and consent of the research study, and therefore cannot be shared. The semi-structured interview guides can be found in the supplemental files.

Declarations**Ethics approval and consent to participate**

The study protocols were reviewed and approved by the Institutional Review Boards of the University of Illinois Chicago (IRB Protocol #2021–1201) and NORC at the University of Chicago (IRB Protocol #23–06–1382). Informed consent was obtained from all participants who consented to the qualitative interviews. Informed consent did not apply to the quantitative data from this project, as it did not include any identifiable information and was not considered human subject research. All procedures were conducted in accordance with the ethical standards of the institutional research committees and with the Declaration of Helsinki of 1964 and its later amendments.

Consent for publication

Not applicable.

Competing interests

The authors declare no competing interests.

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